A Design Space Explorer

app
Performance Predictor
Design Space Explorer
Design Recommendation
Publication Figures

Scene Controls

Parameter mode
- Preset scene
- Custom scene

Custom title
custom structural–dielectric scene

E (dielectric constant) 3.00

d/R 0.1250

Advanced controls

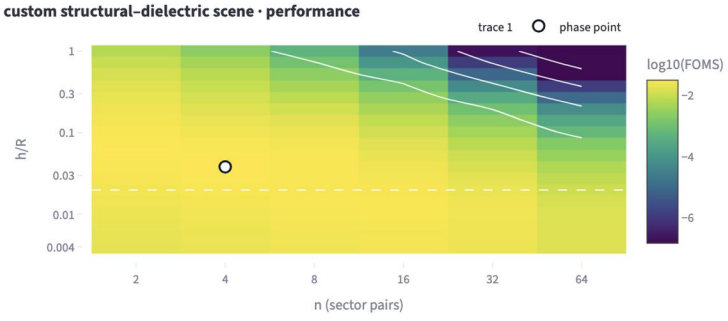
Design Space Explorer

Draw a manuscript-aligned design scene using the current Fig.3/Fig.4 logic. The heatmaps, transition slice, safe-zone overlay, and representative point all reuse the same constants and selection rules as the publication scripts.

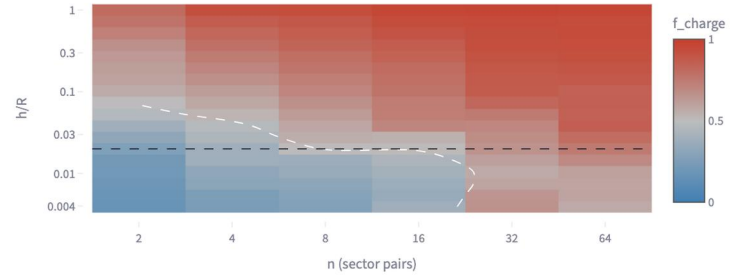
Scene	E / d/R	Phase point	Representative	Safe-zone
custom	3.00 / 0.1250	n=4	h/R=0.0736	Yes

Current scene title: `custom structural-dielectric scene` | dominance threshold = `0.62` | perturbation = `±10%`

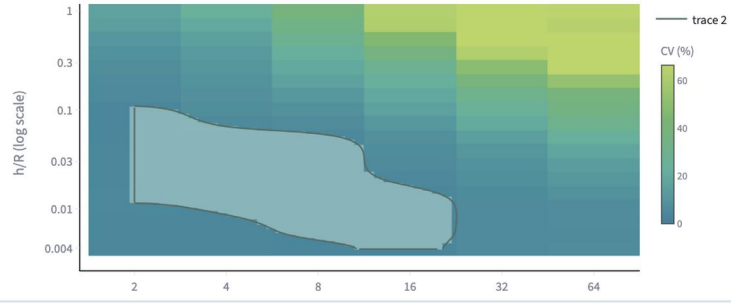
custom structural–dielectric scene · performance



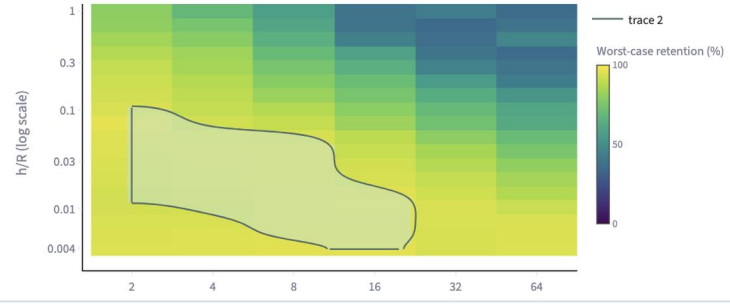
custom structural–dielectric scene · mechanism



custom structural–dielectric scene · robust CV map



custom structural–dielectric scene · worst-case retention



B Performance Predictor

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Design Parameters
- Custom n (OOD)

n (sector pairs) 4

- Custom E (OOD)

E (dielectric constant) 3

dd (gap ratio d/R) 0.1250

hh (thickness ratio h/R) 0.062500

Performance Predictor

Input a structural–dielectric combination to predict TENG metrics and interpret its mechanism-dominance behavior.

Prediction Results

Q_sc (C)	invC_sum (1/F)	FOMS (direct)
1.9632e-09	4.1571e+11	3.0549e-02

Mechanism-dominance

Mixed

f_charge = 0.552
- f_charge > 0.62 → Charge-dominant
- f_charge < 0.38 → Capacitance-dominant
- otherwise → Mixed

This is the same mechanism-dominance interpretation used in the design-space explorer.

Physics Consistency Check

FOMS_direct	FOMS_phys	Relative Deviation
3.0549e-02	3.4069e-02	11.52%

Moderate deviation (11.52%). Predictions may need caution.

The multitask surrogate predicts in scaled log10 space, then restores physical values via scaler.inverse_transform → 10**x.

C Design Recommendation

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Search Controls

Parameter mode
- Preset search space
- Custom search space

Search preset: Fig.4 pooled manuscript pool

Matches the pooled Fig.4 design-candidate space: manuscript n grid, E in [1, 3, 10], d/R fixed at 0.125, and the 18-point h/R grid.

Constraints
Min FOMS 0.00e+0
Max CV (%) 100.00
Min retention (%) 0.00

Advanced controls

Search

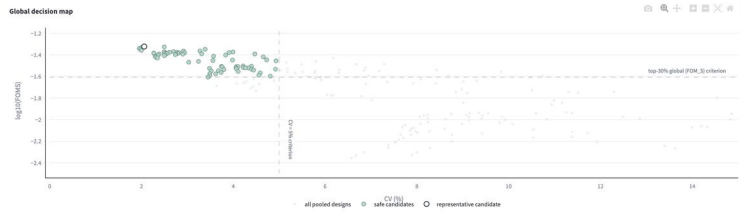
Recommendation Summary

Recommended design (recommend): n=2, E=1.00, d/R=0.1250, h/R=0.053094, FOMS=4.754e-02, CV=2.06%, Retention=97.6%, safe-zone=Yes.

Peak-performance design: n=2, E=1.00, d/R=0.1250, h/R=0.053094, FOMS=4.754e-02, CV=2.06%, Retention=97.6%.

Design Decision Map

Global decision map



This Plotly panel keeps the manuscript Fig.4 styling as closely as possible while adding hover details for every candidate.

Top-10 Designs

Rank	n	E	dd	hh	FOMS	log10_FOMS	Qsc	invC_sum	CV_pct	Retention_pct	foms_top30	safe_zone	recommend_score
1	2	1.00	0.1250	0.053094	4.7542e-02	-1.32	2.5842e-09	8.0016e+11	2.06	97.6	☑	☑	1.195
2	2	1.00	0.1250	0.038317	4.6714e-02	-1.33	2.7342e-09	6.9707e+11	2.00	97.3	☑	☑	1.157
3	2	1.00	0.1250	0.073572	4.7149e-02	-1.33	2.3959e-09	9.2129e+11	2.50	96.7	☑	☑	1.140
4	2	1.00	0.1250	0.027652	4.5714e-02	-1.34	2.8698e-09	6.1516e+11	1.95	97.5	☑	☑	1.118
5	2	1.00	0.1250	0.019956	4.4806e-02	-1.35	3.0102e-09	5.4552e+11	1.97	97.4	☑	☑	1.075